

PRECIOUS RONDILLA
BSIT 4D
WEEK 2 – LAB ACTIVITY

PART 1 : COMPANY PROFILE

Company Name:

TechBox

Nature of Business:

TechBox is a small software development company. It creates websites, mobile apps, and IT solutions for clients.

Company Vision:

To become a trusted company that provides simple, useful, and reliable technology solutions.

Company Mission:

To provide quality software and IT services while keeping client data safe and helping employees work better.

Office Location:

Makati City, Metro Manila, Philippines

Organizational Structure

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

Employee Distribution

- **IT Department:** 5 employees
- **HR Department:** 4 employees
- **Finance Department:** 5 employees
- **Sales Department:** 6 employees

The IT Department will manage the company's computers, network, servers, software, and security.

PART 2 : ENTERPRISE HARDWARE INVENTORY

Asset ID Hardware Quantity Department Purpose

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	20	All Departments	Main computer for employees
HW-002	Laptop	4	IT/Managers	Portable work and troubleshooting
HW-003	Server	1	IT	Store files and run services
HW-004	Router	1	IT	Connect the company to the internet
HW-005	Network Switch	2	IT	Connect computers and devices
HW-006	Printer	2	HR/Finance	Print documents
HW-007	UPS	3	IT	Protect devices from power problems
HW-008	Wireless Access Point	3	All Departments	Provide Wi-Fi
HW-009	NAS Storage	1	IT	Store backup files
HW-010	External Backup Drive	2	IT	Extra backup of important files
HW-011	Monitor	20	All Departments	Display for desktop computers

Computer Specifications:

Desktop Computers

- Intel Core i5 / AMD Ryzen 5
- 16 GB RAM
- 512 GB SSD
- Windows 11 Pro
- Gigabit Ethernet

Laptops

- Intel Core i5 / Ryzen 5
- 16 GB RAM
- 512 GB SSD
- Windows 11 Pro
- Wi-Fi 6

Server

- Intel Xeon or similar server CPU

- 32–64 GB RAM
- 2–4 TB storage
- RAID
- Ubuntu Server
- Dual network ports

These specifications are enough for normal office work, programming, file sharing, and basic IT tasks.

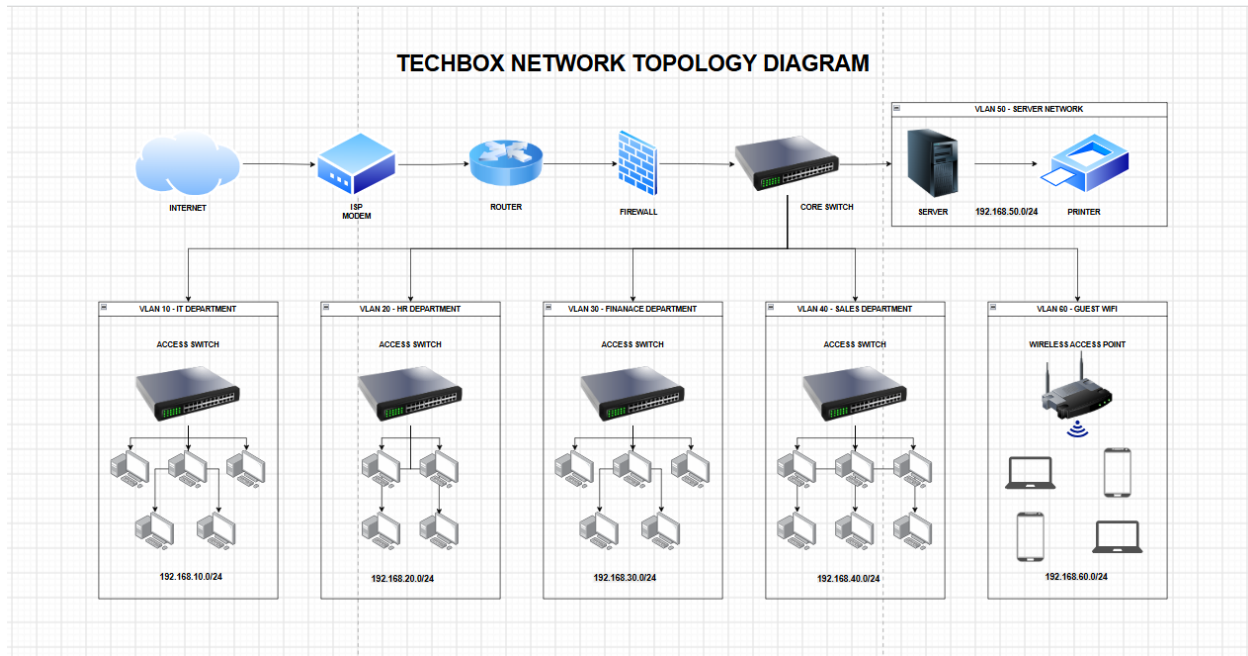
PART 3 : ENTERPRISE SOFTWARE INVENTORY

Software	Version	License	Purpose
Windows 11 Pro	Current	Paid	Main Operating System
Ubuntu Server	LTS	Free	Server operating system
Microsoft Office	Microsoft 365	Subscription	Documents, Excel, PowerPoint, Email
Visual Studio Code	Current	Free	Programming and scripting
Git	Current	Free	Version Control
GitHub Desktop	Current	Free	Manage Git and GitHub
VirtualBox	Current	Free	Create virtual machines
Google Chrome	Current	Free	Web browsing
Microsoft Defender	Built-in	Included	Antivirus and security
AnyDesk	Current	Free/Paid	Remote IT Support
7-Zip	Current	Free	Compress and extract files

PART 4 : ENTERPRISE NETWORK INVENTORY

Asset ID	Network Equipment	Quantity	Purpose
NET-001	ISP Modern	1	Receive internet connection
NET-002	Router	1	Manage network traffic
NET-003	Firewall	1	Protect the company network
NET-004	Managed Switch	2	Connect computers and devices
NET-005	Wireless Access Point	3	Provide Wi-Fi
NET-006	Patch Panel	1	Organize network cables
NET-007	CAT6 Cable	1 lot	Connect network devices
NET-008	RJ45 Connectors	1 lot	Connect network cables

PART 5 : ENTERPRISE NETWORK DIAGRAM



PART 6 : SYSTEM ADMINISTRATION ROLES

1. Helpdesk Technician

Responsibilities

- Help employees with computer problems.
- Install and update software.
- Reset passwords.
- Fix printer problems.
- Fix basic network problems.
- Record user problems.

Skills

- Computer troubleshooting
- Windows
- Basic networking
- Communication
- Problem-solving

Common Tools

- AnyDesk
- Windows tools
- Microsoft 365
- Ticketing system
- Remote support tools

Certifications

- CompTIA A+
- CompTIA Network+
- Google IT Support Certificate

2. Network Administrator

Responsibilities

- Manage routers and switches.
- Configure Wi-Fi.
- Monitor the network.
- Manage IP addresses.
- Fix network problems.
- Maintain network security.

Skills

- TCP/IP
- DNS
- DHCP
- VLANs
- Routing
- Switching
- Network security

Common Tools

- Cisco Packet Tracer
- Wireshark
- PuTTY
- Router and switch management tools

Certifications

- Cisco CCNA
- CompTIA Network+
- Fortinet certifications

3. Linux System Administrator

Responsibilities

- Manage Linux servers.
- Create user accounts.
- Install server software.
- Manage files and permissions.
- Monitor server performance.

- Create backups.

Skills

- Linux commands
- Bash scripting
- File permissions
- Networking
- Server management
- Security

Common Tools

- Ubuntu Server
- Bash
- SSH
- VirtualBox
- Git

Certifications

- Linux+
- LPIC-1
- RHCSA

4. Cloud Administrator

Responsibilities

- Manage cloud servers.
- Manage cloud storage.
- Create cloud users.
- Manage user permissions.
- Monitor cloud services.
- Manage cloud backups.

Skills

- Cloud computing
- Networking
- Security
- Linux/Windows administration
- Backup and recovery

Common Tools

- Microsoft Azure
- Amazon Web Services
- Google Cloud
- PowerShell

- Terraform

Certifications

- Microsoft Azure Administrator
- AWS Certified Cloud Practitioner
- AWS Certified Solutions Architect

The four professionals work as a team to keep TechBox's IT systems working properly. The Helpdesk Technician is usually the first person employees contact when they have a problem. If the problem is related to the network, the Network Administrator handles it. The Linux System Administrator manages the company's Linux server and server services. The Cloud Administrator manages cloud services and cloud backups. They communicate with each other so problems can be solved quickly and company data stays safe.

PART 7 : INFRASTRUCTURE RECOMMENDATIONS

1. Internet Provider

TechBox should choose a reliable business internet provider with:

- At least 500 Mbps speed
- Business customer support
- Good uptime
- Static IP if needed
- Backup internet connection

A business internet plan is recommended because TechBox depends on the internet for daily work.

2. Server Specifications

Recommended server:

- Xeon or similar server CPU
- 32–64 GB RAM
- 2–4 TB storage
- RAID
- Ubuntu Server
- Dual network ports
- UPS protection

The server can be used for file sharing, internal services, and other company needs.

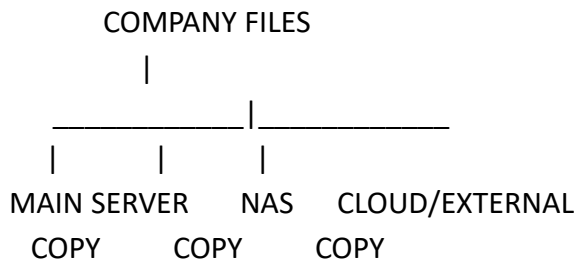
3. Backup Strategy

TechBox should follow the **3-2-1 backup rule**.

- **3 copies** of important data

- **2 different storage types**
- **1 copy in another location or cloud**

Example:



Backups should also be tested regularly to make sure files can be restored.

4. Security Recommendations

TechBox should use:

- Firewall
- Antivirus
- Strong passwords
- Multi-factor authentication
- Regular updates
- User access controls
- Separate department VLANs
- Guest Wi-Fi
- Regular backups
- Security awareness training

These controls can help reduce malware, data leaks, and unauthorized access.

5. Antivirus

All Windows computers should use **Microsoft Defender** or another trusted business antivirus program.

IT staff should make sure:

- Antivirus is active.
- Updates are installed.
- Regular scans are performed.
- Security alerts are checked.

6. Password Policy

TechBox should require:

- Strong passwords.
- At least 12 characters.

- Different passwords for important accounts.
- MFA for important accounts.
- No password sharing.
- Password manager for company accounts.
- Immediate password change if an account may be hacked.

7. Expansion Plan

TechBox currently has 20 employees, but the network should be ready for **40–50 employees** in the future.

The company can:

- Add more switches.
- Add more access points.
- Increase server storage.
- Upgrade internet speed.
- Move some services to the cloud.
- Review security regularly.

PART 8 : PERSONAL REFLECTION

This project helped me understand how important proper planning is before setting up a company's IT infrastructure. I learned that a System Administrator does not only fix computers. A System Administrator also needs to plan hardware, software, networks, security, backups, and user access. I also learned that different IT professionals have different responsibilities, but they still need to work together. Creating the hardware and software inventory helped me understand what equipment and programs a company needs. Designing the network diagram also helped me understand how computers, switches, routers, firewalls, servers, and access points connect.

The most challenging part of this task was designing the network because I needed to think about the needs of all four departments. I also needed to consider security, backups, and future growth. Planning everything before deployment is important because mistakes can become expensive after the system is already installed. I am planning to apply this project before deployment because it gives the company a clear plan for its IT infrastructure. It also helps prevent unnecessary purchases and security problems. This project will help me become a better System Administrator because it improved my planning, networking, documentation, and problem-solving skills. It also gave me a better idea of how IT infrastructure is planned in a real company.